Tensor Categories
Theorem 4.4.1: Absence of Self-Extensions of the Unit

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Fixed source and scope. *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, 2015, author final copy.
`https://math.mit.edu/~etingof/egnobookfinal.pdf`.
The retained PDF has 362 physical pages; its metadata record creation on May 13, 2015 and modification on July 29, 2015. The primary is the statement on physical 87 (printed 70) and its complete proof on physical 88 (printed 71). Corollary 4.4.2 and Remark 4.4.3 are uncounted context.

Chapter 4 opens on physical 82 (printed 65) with "Let k be an algebraically closed field." Section 4.4 adds characteristic zero. The theorem assumes a finite *ring* category and a simple unit; it does not assume left or right duals, rigidity, symmetry, semisimplicity, or a fiber functor. The original "(right) action" and "homomorphism" wording, "etc." and "We can now proceed further" are retained. The passage to higher tensor powers and the final finite-character sentence are not expanded into a new proof.

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Definition context (uncounted)

Definition 4.2.3 (physical 83, printed 66) reads: a *multiring category* over k is a locally finite k-linear abelian monoidal category $\mathcal{C}$ with bilinear and biexact tensor product. If in addition $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) = k$, we will call $\mathcal{C}$ a *ring category*. No duals occur in this definition.

Definition 1.8.5 (physical 26, printed 9) says that a k-linear abelian category $\mathcal{C}$ is *finite* if it is equivalent to the category A-mod of finite dimensional modules over a finite dimensional k-algebra A. Definition 1.8.6 gives the equivalent conditions: finite dimensional Hom spaces, finite length of every object, enough projectives (every simple object has a projective cover), and finitely many isomorphism classes of simple objects. The first two conditions alone define *locally finite* (Definition 1.8.1). The equivalence of these two definitions, discussed on physical 27 (printed 10), is an explicit prior input; its proof is not reprinted.

Definition 1.6.5 (physical 23, printed 6) calls P projective if $\mathrm{Hom}_{\mathcal{C}}(P, -)$ is exact. Under the finite-length condition, Definition 1.6.6 calls $p : P(X) \to X$ a *projective cover* if $P(X)$ is projective, p is an epimorphism, and for every epimorphism $g : P \to X$ from a projective object there is an epimorphism $h : P \to P(X)$ such that $ph = g$. Such a cover, when it exists, is unique up to a non-unique isomorphism.

Definition 1.4.2 (physical 21, printed 4) defines morphisms between short exact sequences $0 \to X \to Z \to Y \to 0$ and $0 \to X \to Z' \to Y \to 0$ as maps $Z \to Z'$ restricting to the identity on X and inducing the identity on Y. The set of these sequences up to isomorphism is $\mathrm{Ext}^1(Y, X)$, the set of extensions of Y by X. Its addition and identification with derived Ext are standard prior background (Section 1.4 and physical 24, printed 7); their proofs are not reprinted. The notation A^* in the proof below means the vector-space dual of the finite dimensional algebra A. The ordinary symbol Vec denotes finite dimensional k-vector spaces.

4.4. Absence of self-extensions of the unit object

Assume that k has characteristic 0.

Theorem 4.4.1. *Let $\mathcal{C}$ be a finite ring category over k with simple object $\mathbf{1}$. Then* $\mathrm{Ext}^1(\mathbf{1}, \mathbf{1}) = 0$.

Proof. Assume the contrary, and suppose that V is a nontrivial extension of $\mathbf{1}$ by itself. Let P be the projective cover of $\mathbf{1}$. Then $\mathrm{Hom}(P, V)$ is a 2-dimensional space, with a filtration induced by the filtration on V, and both quotients naturally isomorphic to $E := \mathrm{Hom}(P, \mathbf{1})$. Let v_0, v_1 be a basis of $\mathrm{Hom}(P, V)$ compatible to the filtration, i.e., v_0 spans the 1-dimensional subspace defined by the filtration. Let $A = \mathrm{End}(P)$ (this is a finite dimensional algebra). Let $\varepsilon : A \to k$ be the character defined by the (right) action of A on E. Then the matrix of $a \in A$ in the basis v_0, v_1 has the form

$$[a]_1 = \begin{pmatrix} \varepsilon(a) & \chi_1(a) \\ 0 & \varepsilon(a) \end{pmatrix} \tag{4.5}$$

where $\chi_1 \in A^*$ is nonzero. Since $a \mapsto [a]_1$ is a homomorphism, χ_1 is a derivation: $\chi_1(xy) = \chi_1(x)\varepsilon(y) + \varepsilon(x)\chi_1(y)$.

Now consider the representation $V \otimes V$. Using the exactness of the tensor product, we see that the space $\mathrm{Hom}(P, V \otimes V)$ is 4-dimensional, and has a 3-step filtration, with successive quotients $E, E \oplus E, E$, and basis $v_{00}; v_{01}, v_{10}; v_{11}$, consistent with this filtration. The matrix of $a \in \mathrm{End}(P)$ in this basis is

$$[a]_2 = \begin{pmatrix} \varepsilon(a) & \chi_1(a) & \chi_1(a) & \chi_2(a) \\ 0 & \varepsilon(a) & 0 & \chi_1(a) \\ 0 & 0 & \varepsilon(a) & \chi_1(a) \\ 0 & 0 & 0 & \varepsilon(a) \end{pmatrix}. \tag{4.6}$$

Since $a \mapsto [a]_2$ is a homomorphism, we find

$$\chi_2(ab) = \varepsilon(a)\chi_2(b) + \chi_2(a)\varepsilon(b) + 2\chi_1(a)\chi_1(b), \qquad a, b \in A.$$

We can now proceed further (i.e., consider $V \otimes V \otimes V$ etc.) and define for every positive n, a linear function $\chi_n \in A^*$ which satisfies the equation

$$\chi_n(ab) = \sum_{j=0}^{n} \binom{n}{j} \chi_j(a)\chi_{n-j}(b),$$

where $\chi_0 = \varepsilon$. Thus for any $s \in k$, we can define $\varphi_s : A \to k((t))$ by $\varphi_s(a) = \sum_{m \geq 0} \chi_m(a) s^m t^m / m!$ (this is where we use that k has characteristic 0) and we find that φ_s is a family of pairwise distinct homomorphisms. This is a contradiction, as A is a finite dimensional algebra and can have only finitely many 1-dimensional representations over any extension field. We are done. □

Consequence and scope contrast (uncounted)

Physical 88, printed 71; complete printed passages.

Corollary 4.4.2. If a finite ring category $\mathcal{C}$ over a field of characteristic zero has a unique simple object $\mathbf{1}$, then $\mathcal{C}$ is equivalent to the category Vec.

Remark 4.4.3. The conclusion of Corollary 4.4.2 does not hold when the ground field has a positive characteristic. We give a counterexample: the category $\mathrm{Rep}(\mathbb{Z}/p\mathbb{Z})$ over a field of characteristic p.

Separate 2026 corrections and source credit

The six-page author correction document is dated February 14, 2026: `https://math.mit.edu/~etingof/tensbookcor.pdf`. No entry in its Chapter 4 list changes Theorem 4.4.1 or its displayed proof. The physical-page-3 entries change the reference in the proof of Proposition 4.2.8, interchange "monomorphisms" and "epimorphisms" in Corollary 4.3.9, and require finiteness in the paragraph before Example 4.5.5. Those passages are not imported as hypotheses or silently substituted into this proof. The primary's finite-ring and characteristic-zero hypotheses already appear in the 2015 source.

The four book authors supply the exposition transcribed here. Note 4.4 on physical 105 (printed 88) states: "The results on absence of self-extensions are taken from [EtO1] (see Theorem 2.17)." The cached book bibliography on physical 345 (printed 328) gives:

> [EtO1] P. Etingof and V. Ostrik. *Finite tensor categories.* Moscow Math. Journal 4 (2004), 627–654.

This is the book's historical credit, not a claim that the separate paper was obtained or that its other results are included. No new derivation of the higher-χ_n construction, finite-character assertion, or other author shorthand is supplied.